Name ____ Date ___________ Period _____

Domain and Range: All Three Notations

Algebra II | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write the domain and range of a function from a graph, a table, or a set of ordered pairs using a verbal description, inequality notation, set notation, and interval notation.

Vocabulary

Domain — the set of all _______________ values, the ____-values.

Range — the set of all _______________ values, the ____-values.

Set notation — uses ___________________ with the bar | , read "such that."

Interval notation — uses _________ if an endpoint is included, _________ if it is excluded.

Continuous — the graph is _________________, with no breaks or gaps.

Discrete — the graph is made of _________________ points.

Part 1 — From a Set of Ordered Pairs

Least to greatest, and each value written only once.

Example 1. { (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) }

Domain: { _____________________________________ }

Range: { _____________________________________ }

You Try 1. { (-5, 3), (-1, 0), (2, 3), (6, -8) }

Domain: { _____________________________________ }

Range: { _____________________________________ }

Part 2 — From a Table

Example 2.

x	-3	-1	0	2	5
f(x)	9	1	0	4	25

Domain: { _____________________________________ } Range: { _________________________________ }

You Try 2.

x	-2	0	1	4
f(x)	7	7	-3	10

Domain: { _____________________________________ } Range: { _________________________________ }

Part 3 — The Three Notations

The first row is done for you. Fill in the rest.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	$-\infty < x < \infty$	$\{ x \mid x \in \mathbb{R} \}$	$(-\infty, \infty)$
all reals greater than or equal to 0	$x \geq 0$	____________________	____________

all reals less than 5	x < 5	____________________	____________
all reals from -3 to 3, including both	$-3 \le x \le 3$	____________________	____________
just the numbers -2, 0, and 4	*(none)*	____________________	____________

Key idea: bracket [ ] = endpoint _______________. Parenthesis ( ) = endpoint _______________. ∞ always takes a _______________.

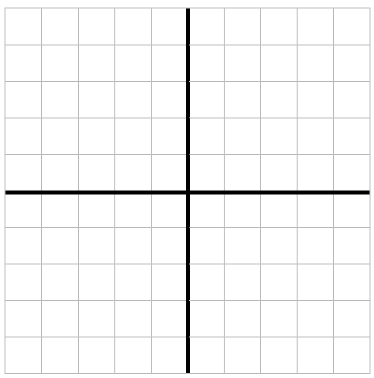
You Try 3. all reals greater than -8 → inequality _________________ interval _________________

You Try 4. $y \le 12$ → in words ______________________________ interval _______________

Part 4 — From a Graph

Each square is 1 unit. The axes are the two heavy lines, and both grids run -5 to 5.

A. Discrete. Plot (-4, 2), (-2, -3), (1, 4), (3, -1).

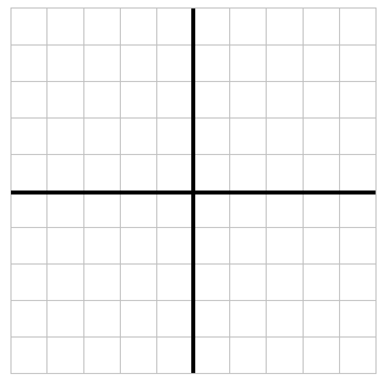


Domain (set): _________________

Range (set): _________________

Continuous or discrete? _______________

B. Continuous. Graph y = x + 1 from x = -4 to x = 3.



Domain (inequality): _________________

Domain (interval): _________________

Range (interval): _________________

Part 5 — Continuous or Discrete?

Write continuous or discrete.

1. The number of students in each class period ______________________
2. The outside temperature over one full day ______________________
3. The total cost of n concert tickets at $45 each ______________________

Exit Ticket

Given { (1, 5), (3, 5), (4, -2) } write the domain and range in set notation, then say whether interval notation could be used.

Domain ________________________ Range ________________________

Could interval notation be used? _________________________________